

Validity of the Open Card Sorting Method for Producing Website Information Structures

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ABSTRACT

When it comes to creating website information structures, open card sorting is the main approach used. However, scientific research for the method's validity is lacking. This paper explores the validity of open card sorting for website structural design. To this end, participants first performed an open card sort for the redesign of a website with usability issues in its information structure. Next, a within-subjects user testing study was performed to compare two functional prototypes that differed only in their structure: one replicated the existing website structure, whereas the other implemented the structure produced by card sort data analysis. Results showed that participants using the redesigned structure had significantly better usability metrics (first click success rate, task time, SEQ score and SUS score) compared to when interacting with the existing structure. These findings provide support for the validity of the open card sorting method for the design of website information structures.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); HCI design and evaluation methods; Empirical studies in HCI.

KEYWORDS

Card sorting, Information architecture, Validity, Method evaluation

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1 INTRODUCTION

Card sorting is a method used to understand how participants would organize and structure content that makes sense to them. It can be used to provide insights into participants' mental models in a variety of situations, such as health professional's perspectives on self-tracking data for mental wellness in clinical contexts [23], children's

perceptions of colors, symbols and words in cybersecurity warnings [15], children's categorization of mobile games in app stores [7], web designers' programming knowledge and learning strategies [11], experts' perspectives on organizing a collection of haptic devices [36], designers' groupings of guidelines for the design of robot teleoperation [1], DVD menus [16] and age-friendly websites [43]. Card sorting has been also used to validate AutoCardSorter, a tool for automated grouping of content in interactive systems [17, 18, 20].

However, the most common use of card sorting is to support the design or evaluation of website Information Architecture (IA) [2, 33, 37]. The IA of an interactive system specifies how its content is structured and labeled [33]. A card sort requires participants to sort content elements (cards) into categories. There are various variants of the method. In an open card sort, participants sort the cards into their own categories and give each one a name. In a closed card sort, participants sort the given cards into a set of predetermined labeled categories. The hybrid variant [14] combines open and closed card sorting, allowing users to add new categories to predefined ones. In the Modified-Delphi version [30], a seed participant conducts an open card sort, followed by subsequent participants commenting on and modifying the prior participants' sort until no substantial modifications occur.

Open card sorting is the most widely-used variant [37, 41]. The object of this paper is to investigate the validity of the open card sorting method for producing website information structures. Previous research has looked into a variety of methodological aspects related to executing an open card sort and interpreting the collected data. In terms of sample size, 15 to 20 participants has been shown to be sufficient [26, 39]. Participants' characteristics, such as sense of direction [42] and self-efficacy [22], have been found to significantly affect the obtained information structures.

In addition, the time it takes participants to finish an open card sort is largely impacted by the quantity of cards to be sorted [14]. In specific, it can take from 20 minutes for 30 cards to 60 minutes for 100 cards [14]. According to Spencer [37], the number of cards to sort should be neither less than 30 because forming groups might be problematic, nor more than 100 because the participants may become tired or confused. When sorting lots of cards, Tullis and Wood [40] recommend having each participant sort only 60% of the whole set. They found that if 30-40 participants are involved in the sort then the obtained results are quite close to sorting the entire set of cards.

Card sorting can be done manually with actual cards or electronically with software. There is no difference in the obtained results from manual and software-mediated card sorting, according to studies [13, 31]. The usability of card sorting tools has been also evaluated [8], with researchers preferring WebSort and end-users

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selecting OpenSort. However, more recent research is required towards this direction as the UI and functionality of most tools in [8] that are still available today have changed and new card sort tools have emerged, such as the free and open source CardSorter [24].

Regarding analysis of open card sort data, examining the same data using different approaches results in varied IAs [25]. Righi and colleagues [32] discuss how to analyze data from open card sorting activities and what to look for. According to Spencer [37], data can be explored by exploratory (qualitative) or statistical (quantitative) analysis, or a mix of the two. Exploratory analysis helps to uncover patterns in the collected data that illustrate how participants think about the cards and groups by using tabulations [37] or graph visualizations [30] of the data. Statistical analysis of open card sort data employs a variety of approaches, including hierarchical cluster analysis [2, 10, 31, 37], k-means clustering [28, 37], multidimensional scaling [2, 28, 37], and factor analysis [6]. The most often used method is hierarchical cluster analysis [14, 25, 32, 41]. A dendrogram, a tree diagram that shows a hierarchy of groupings based on the dissimilarity of content items, is the principal result of hierarchical cluster analysis. This dendrogram can reveal information about content arrangement and menu structure.

Open card sorting seems to be a common method to facilitate the creation of website IAs. Various parameters related to the methodology of conducting and analyzing open card sort data have been investigated in the literature. By contrast, the validity and reliability of the method remain largely unexplored. Recent research [19, 29] supports the cross-study and test-retest reliability of open card sorts. However, we did not find any research providing empirical evidence on the validity of the open card sorting method. This paper addresses the following research question: Is the open card sorting method a valid method for producing usable website information structures?

In an attempt to answer this question, we conducted a two-phase study. In the first phase, we chose a website with IA-related usability issues and performed an open card sort with content items selected from this website. In the second phase, we implemented two interactive prototypes, one with the existing website structure and one with the structure produced by analysis of the collected open card sort data, and conducted a within-subjects user testing study to compare usability metrics for these prototypes. The methodology and results of these two studies are reported in the following two sections, and the paper concludes with the findings and future research directions.

2 CARD SORTING STUDY

2.1 Participants

The open card sort involved 20 participants, 11 males and 9 females, with a mean age of 30.5 years ($SD=9.7$). Open card sorts require a minimum of 15 users to produce reliable data [26, 39], therefore our sample size is appropriate. All of the participants were native speakers of Greek, the language used in the cards.

2.2 Website selection and cards

We examined various websites of various domains (e.g., academic websites, local government organizations, entertainment) with the aim to identify one with IA-related usability issues so that it could

benefit by a redesign based on open card sort data analysis. A total of 10 candidate websites were evaluated in a pilot user testing study involving 5 participants, 3 females and 2 males. The pilot study collected data on participants' task success, time on task, Single Ease Question (SEQ) [35] scores and System Usability Scale (SUS) [5] scores. The website with the worst performance in the collected usability metrics was selected for the card sort. This was a website of a local government organization.

Following Spencer's [37] recommendations, a total of 49 cards were chosen from the website. Examples of the selected cards are the following: "Organization of the municipality", "Museums", "The park of our city", "Ski center", "Carnival", "Register of volunteers", "Press releases", "Announcements", and "Contact form". The chosen cards were pilot-tested to verify that they were similar enough to propose possible groups but not too similar to lead participants to a certain outcome. The card labels and descriptions were straightforward to read and reflective of the website domain, according to the pilot-testers.

2.3 Instruments and procedures

The OptimalSort by Optimal Workshop (<https://www.optimalworkshop.com>) online card sorting tool was used to mediate the open card sort remotely. Participants were provided with the link to the OptimalSort project. This project first presented a brief questionnaire capturing participants' gender and age. Next, participants completed the open card sort individually. The cards to sort were set to appear in random order in the OptimalSort environment. Participants could optionally leave a comment before finalizing their submission. Participants' comments concerned either their overall experience with the open card sort or clarifications regarding their taxonomy. The collected data were analyzed using the functionality provided by OptimalSort. The data analysis method is delineated in the following section.

2.4 Data analysis and results

The data from the open card sort were analyzed using a combination of exploratory and statistical methods [37]. Spencer's spreadsheet template [37] was used to conduct the exploratory analysis. First, the data from the open card sort were inserted into the spreadsheet. This automatically filled up a cards $\times$ participants matrix, with the label of the group in which the user grouped the card in each cell. Then, for each group, a standardized label was chosen. The standardized label was the one that "had been used by most participants or represented the idea most clearly" [37]. This automatically generated a matrix that visualized the categorization of cards using standardized labels. This matrix provides insights on the groups created by participants and the labels they used for them.

Statistical analysis was also used to help us construct the final categories. To this end, the analysis features provided by OptimalSort were used. In specific, we used the best-merge method [25] dendrogram (Figure 1) and the 3D cluster view (Figure 2). In addition, we performed factor analysis using IBM SPSS 25 to determine the statistically optimal number of categories.

The data analysis process resulted in the redesigned information structure, that is the one based on analysis of the open card sorting data. This structure was compared to the existing structure of the

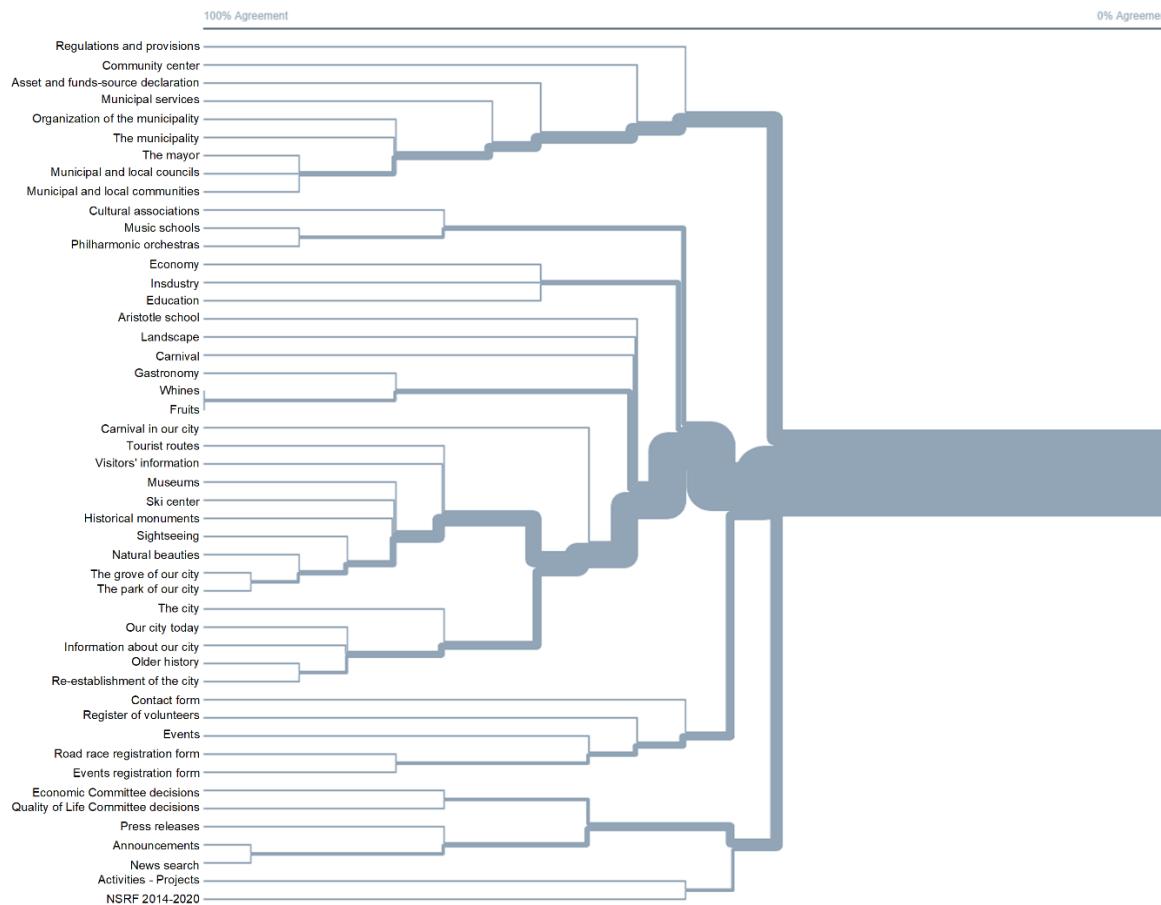


Figure 1: Best-merge method dendrogram for our open card sort dataset in OptimalSort. Items were made available in participants' native language (Greek) but are translated into English for presentation purposes.

government website through a within-subjects user testing study reported in the next section.

3 USER TESTING STUDY

3.1 Participants

The within-subjects user testing involved 30 participants, 20 females and 10 males, with a mean age of 27.2 years ($SD=3.9$). All participants were native speakers of Greek, the language used in the provided prototypes and questionnaires. None of the participants was involved in the open card sort study elaborated in the previous section.

3.2 Prototypes

Two web-based functional prototypes were produced for the government website. Both prototypes presented a top navigation menu and had the same look and feel. The first prototype replicated the IA of the existing website, whereas the second one implemented the IA produced by the analysis of the open card sorting study data.

3.3 Instruments and procedures

Videoconferencing and remote control of the facilitator's laptop were used to conduct this study remotely. Participants were given four tasks to complete in both functional prototypes of the government website (existing IA, redesigned IA). The tasks asked participants to find: a) the latest announcements of the municipality, b) what services are provided by the municipality to its citizens, c) contact information of the community center, and d) historical information about the industry of the municipal area. We chose content items from the two IA versions that were grouped under different category names. The order of both the prototype version and tasks was randomized per user to minimize order effects. A custom-built computer script recorded whether participants clicked on the correct category on their first try (first click success) and how long they spent on each task (time on task). Participants were informed that they could give up a task at any time.

After each task, the SEQ [35] was presented to participants. SEQ measures task difficulty on a scale from 1 (very difficult) to 7 (very easy). It has been found [35] to perform as well or better than other



Figure 2: 3D cluster view for our open card sort dataset in OptimalSort. Items were made available in participants' native language (Greek) but are translated into English for presentation purposes.

Table 1: Descriptive statistics of the dependent variables measured in this study.

Website IA version	Variable	Mean	Median	SD	95% C.I.
Existing	First click success (%)	20.0	25.0	17.9	(13.3, 26.7)
Redesigned	First click success (%)	72.5	75.0	19.0	(65.4, 79.6)
Existing	Time on task (sec)	32.2	31.2	11.2	(28.0, 36.4)
Redesigned	Time on task (sec)	20.0	17.0	8.2	(17.0, 23.1)
Existing	SEQ score (1-7)	4.1	3.9	1.3	(3.6, 4.6)
Redesigned	SEQ score (1-7)	6.6	6.7	0.5	(6.4, 6.8)
Existing	SUS score (0-100)	57.4	55.0	27.4	(47.2, 67.7)
Redesigned	SUS score (0-100)	91.8	97.5	11.1	(87.7, 96.0)

measures of task difficulty like the Subjective Mental Effort Questionnaire (SMEQ) and the Usability Magnitude Estimation (UME). After performing all tasks per prototype version, participants completed the SUS [5]. SUS measures perceived usability on a scale from 0 to 100, where higher scores indicate better system usability. It uses ten items, five positive and five negative statements, to be scored on a scale from 1 (strongly disagree) to 5 (strongly agree). SUS is a valid and reliable instrument for measuring perceived usability [3, 38]. In our research, we employed the standardized Greek version of SUS [21, 27]. The Greek SUS has been found to be reliable in a dataset of 436 surveys; Cronbach's $\alpha = 0.81$ [27]. In our dataset, it was also found to be reliable; Existing IA: Cronbach's $\alpha = 0.94$, Redesigned IA: Cronbach's $\alpha = 0.83$.

The Google Forms service was used to create and distribute SEQ and SUS. The collected data were organized using Microsoft Excel 2016 and were analyzed using IBM SPSS Statistics 25.

3.4 Results

Table 1 presents descriptive statistics of the dependent variables of this study.

Composite scores over all four tasks were produced for first click success, time on task and SEQ score. First click success rate was calculated as the percentage of tasks in which participants found the correct item with their first click. Task time was calculated as the average of all four tasks, regardless of whether a task was successfully completed or not. As participants were offered the choice to stop a task, producing a composite score incorporating all task durations offers a more realistic picture of total interaction efficiency [2]. SEQ ratings were averaged across all four tasks to produce a composite score.

In all subsequent statistical analyses, the effect size r was calculated according to the formulas reported in [12].

3.4.1 First click success. Figure 3 shows the participants' mean first click success for the prototype with the existing IA and the one with the redesigned IA based on the open card sort data analysis. A Shapiro-Wilk test found that the distribution of the differences in the first click success for the existing IA and the redesigned IA deviated significantly from a normal distribution; $W(30)=0.918$, $p=0.015$. Thus, a nonparametric test was used to compare participants' first click success between the two conditions. A two-tailed

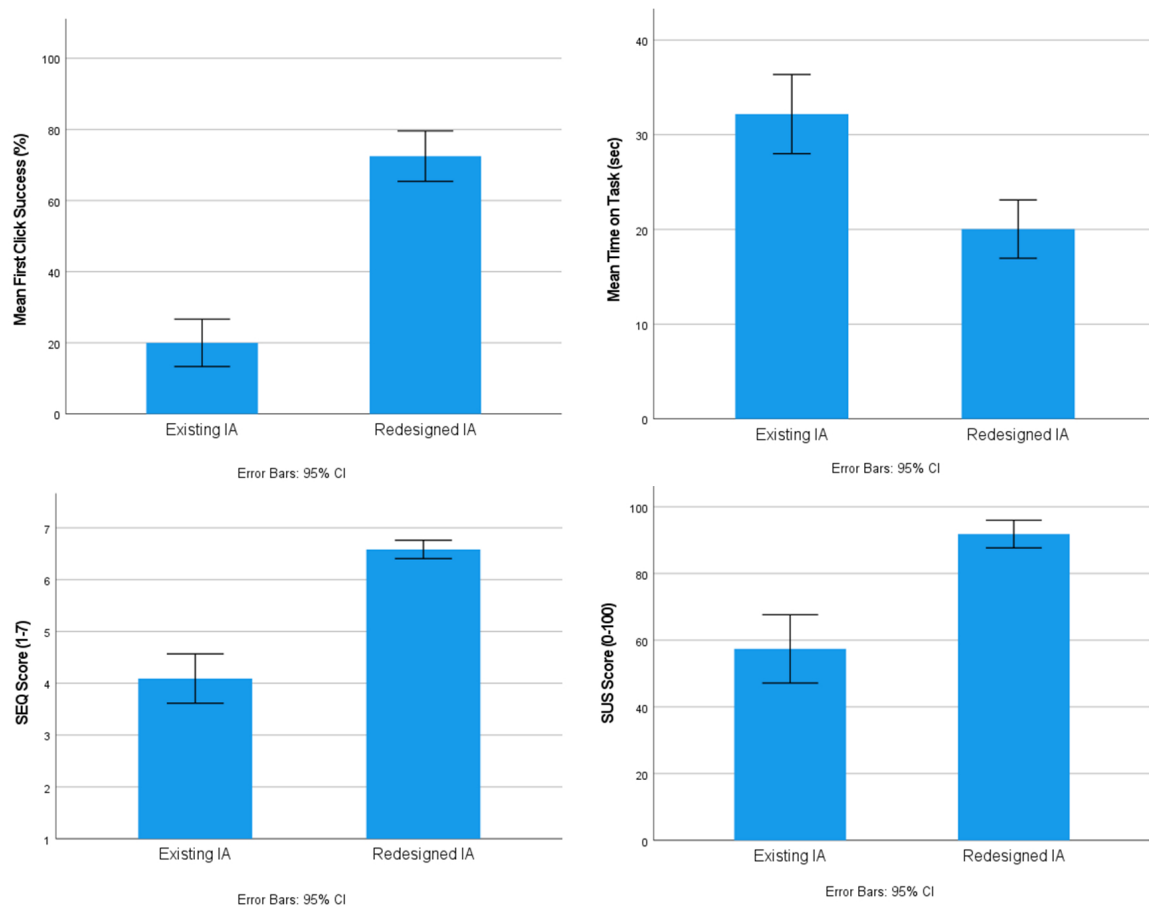


Figure 3: Bar charts of the dependent variables collected in the user testing study: mean participants' first click success (top-left), time on task (top-right), SEQ score (bottom-left), and SUS score (bottom-right). Error bars represent the 95% confidence interval.

Wilcoxon signed-rank test found that participants were significantly more successful with their first click in the redesigned IA (Mdn=75.0%) compared to the existing one (Mdn=25.0%); $z=4.763$, $p<0.001$, $r=0.615$.

3.4.2 Time on task. Figure 3 displays the participants' time on task per IA version. Shapiro-Wilk analysis showed that the assumption of normality was not violated for the differences in the task times of the two conditions; $W(30)=0.958$, $p=0.275$. Thus, a parametric test was used to compare participants' time on task between the existing IA and the redesigned IA based on open card sort data analysis. A two-tailed dependent t-test found a significant effect of the website IA version on participants' time on task; $t(29)=5.575$, $p<0.001$, $r=0.719$. Participants were significantly faster when navigating on the website with the redesigned IA ($M=20.0$ sec, $SD=8.2$ sec) than when they did so on the website with the existing IA ($M=32.2$ sec, $SD=11.2$ sec).

3.4.3 Single Ease Question score. Figure 3 illustrates the participants' SEQ score per website version. A Shapiro-Wilk test found that the distribution of the differences in the SEQ score for the

existing IA and the redesigned IA deviated significantly from a normal distribution; $W(30)=0.918$, $p=0.024$. Thus, a nonparametric two-tailed Wilcoxon signed-rank was used. It found that there was a significant difference between participants' SEQ score in the existing IA (Mdn=3.9) and the redesigned IA (Mdn=6.7); $z=4.709$, $p<0.001$, $r=0.608$. The participants interacting with the IA that was redesigned based on the open card sort data analysis found the tasks significantly easier than when interacting with the existing IA. In a dataset of over 400 tasks and 10000 users [34], the average SEQ score is between 5.3 and 5.6. Thus, the same tasks were perceived as rather difficult in the existing IA and rather easy in the redesigned IA.

3.4.4 System Usability Scale score. Figure 3 shows the participants' SUS score per website IA. Reliability analysis found a high internal reliability of SUS for our dataset; Existing IA: Cronbach's $\alpha=0.938$, Redesigned IA: Cronbach's $\alpha=0.827$. Shapiro-Wilk analysis showed that the assumption of normality was not violated for the differences in the SUS score of the two conditions; $W(30)=0.966$, $p=0.442$. Thus, a parametric test was used to compare participants' SUS score

between the existing IA and the redesigned IA. A two-tailed dependent t-test found a significant effect of the website IA version on participants' SUS score; $t(29)=7.356$, $p<0.001$, $r=0.807$. Participants' SUS score was significantly higher for the IA that was redesigned based on the open card sort data analysis ($M=91.8$, $SD=11.1$) than for the existing IA ($M=57.4$, $SD=27.4$). According to Bangor and colleagues [4], the existing IA is characterized as "OK to Good" and the redesigned IA is "Best Imaginable" in terms of perceived usability as measured by SUS.

4 CONCLUSIONS

The goal of this paper was to assess the open card sorting method's validity for supporting the IA design of websites. To that goal, two studies were conducted. First, 20 participants completed an open card sort for a local government organization's website, which had usability issues in its information structure. Next, a separate group of 30 participants was involved in a within-subjects user testing study that compared interaction performance and self-reported metrics for two functional prototypes: one with the existing IA and one with a redesigned IA produced by state-of-the-art analysis of the open card sort data.

Results showed that participants using the prototype with the redesigned IA made significantly more correct first clicks, required significantly less time to find content items, characterized the tasks as significantly easier and rated its perceived usability as significantly higher compared to when they interacted with the prototype implementing the existing IA. In all comparisons, an effect size from $r=0.608$ to $r=0.807$ was found, which shows a large effect [9].

In conclusion, this paper found empirical evidence for the validity of the open card sorting method when used to support the information structure design of websites. This finding is particularly important because it gives scientific justification for a widely used method in HCI.

One limitation of this study is that it only looked at one website domain (i.e., government websites), hence more research with different domains of websites is needed to establish that the findings are generalizable. Additional studies with varying number of cards are also required. Furthermore, this research selected a website with poor usability and showed that its usability is improved after the redesign of its IA based on open card sorting. It would be interesting to explore whether websites with good usability can also benefit from open card sorting.

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